

Method	Prompts	Global Dice	Macro Dice	mIoU	Boundary F1
SAM	(1+1)	0.4860	0.4649	0.3094	0.1369
SAM-Med2D		0.4219	0.4318	0.2979	0.1454
WSI-SAM		0.4154	0.4064	0.2653	0.1353
FewClick		0.7522	0.7418	0.5938	0.2971
SAM	(3+1)	0.5903	0.5805	0.4143	0.1763
SAM-Med2D		0.5178	0.5323	0.5089	0.1680
WSI-SAM		0.5623	0.5608	0.3976	0.1676
FewClick		0.8164	0.8159	0.6913	0.3324
SAM	(5+1)	0.6290	0.6243	0.4585	0.1956
SAM-Med2D		0.6050	0.6149	0.4706	0.1791
WSI-SAM		0.6041	0.6057	0.4397	0.1764
FewClick		0.8425	0.8406	0.7178	0.3229

Table 1: Patch-level segmentation under matched episodes and prompt coordinates. Best results within each prompt budget are in bold.

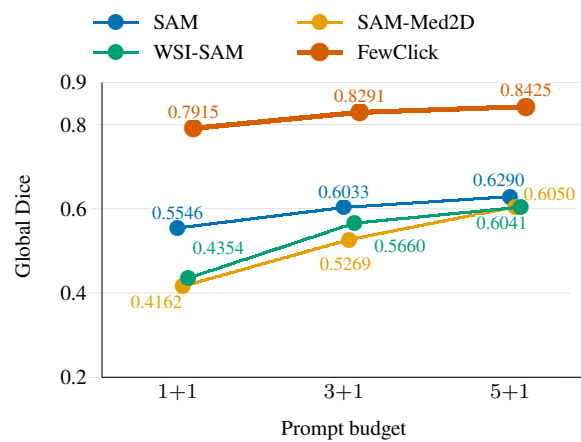


Figure 2: Global Dice versus prompt budget on the common 885 occurrences (819 unique patches). Methods share targets, masks, negative points, and prefixes of the same frozen positive-point sequence.

Method	Macro Dice	Macro mIoU	Macro Prec.	Macro Recall	Micro Dice
SAM	0.3352	0.2344	0.2372	0.9628	0.5424
SAM-Med2D	0.3478	0.2276	0.2987	0.7152	0.4555
WSI-SAM	0.4001	0.2871	0.2925	0.9533	0.6257
FewClick	0.7495	0.6383	0.7456	0.7676	0.9237

Table 2: Oracle-assisted whole-slide results on five WSIs and 12 tissues. FewClick reports the best of five GT-selected candidates per WSI–tissue pair; each SAM baseline uses the revised context-aware, GT-guided patchwise protocol. These attainable-protocol results complement the matched patch-level comparison.

Tissue	FewClick	SAM	SAM-Med.	WSI-SAM
Tumor epi.	0.8859	0.5695	0.5468	0.6186
Tumor stroma	0.8443	0.4067	0.4733	0.4976
Background	0.9776	0.8263	0.4573	0.8481
Necrosis	0.5590	0.0735	0.1484	0.1089
Normal gland	0.6963	0.3060	0.3531	0.3652
Normal stroma	0.6430	0.2082	0.2722	0.2719
Submucosa/serosa	0.8500	0.4423	0.5080	0.5493
Muscle	0.8611	0.5106	0.5643	0.6001
Lymph. aggregate	0.7937	0.2064	0.3625	0.3231
Mucus	0.6590	0.1066	0.1526	0.1522
Adipose	0.6820	0.3374	0.2746	0.4139
Blood	0.5419	0.0292	0.0604	0.0529

Table 3: Per-class whole-slide Dice under the oracle-assisted protocols described in the text. Best results are in bold.

Variant	Macro Dice	Micro Dice	Boundary F1	Δ Macro
FewClick	0.7413	0.8138	0.3142	–
Fine scale only	0.6384	0.6929	0.2630	–0.1029
No cell fusion	0.5885	0.6543	0.2471	–0.1528
Fixed SLIC regions	0.5650	0.6482	0.1475	–0.1763
Mean prompt prototypes	0.7215	0.7970	0.3062	–0.0199

Table 4: Component ablations on 4,000 fixed validation episodes. Δ is the macro-Dice change from FewClick.